

Multiple Choice

1. **Answer:** A

Options: A) asteroids would orbit with a period half that of Jupiter; B) asteroids would orbit with a period twice that of Jupiter; C) asteroids would orbit with a period twice that of Mars; D) A and B

Note: Correct option: A - asteroids would orbit with a period half that of Jupiter

2. **Answer:** D

Options: A) fast rotation only; B) a rocky mantle only; C) a molten metallic core only; D) both a molten metallic core and reasonably fast rotation

Note: Correct option: D - both a molten metallic core and reasonably fast rotation

3. **Answer:** C

Options: A) a New Horizons flyby in the 1990 s; B) Hubble Space Telescope images that resolved Pluto's disk; C) brightness measurements made during mutual eclipses of Pluto and Charon; D) radar observations made by the Arecibo telescope

Note: Correct option: C - brightness measurements made during mutual eclipses of Pluto and Charon

4. **Answer:** C

Options: A) a composition dominated by hydrogen and helium; B) the presence of belts zones and storms; C) an equatorial wind speed of more than 900 miles per hour; D) significant "shear" between bands of circulation at different latitudes

Note: Correct option: C - an equatorial wind speed of more than 900 miles per hour

5. **Answer:** C

Options: A) the entire planets are made mostly of metal.; B) metals condensed first in the solar nebula and the rocks then accreted around them.; C) metals sank to the center during a time when the interiors were molten throughout.; D) radioactivity created metals in the core from the decay of uranium.

Note: Correct option: C - metals sank to the center during a time when the interiors were molten throughout.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '11'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'the rigid rocky material of the crust and uppermost portion of the mantle.'

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'You see the star's blackbody spectrum with absorption lines due to hydrogen.'

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Jupiter's greater mass compresses it more thus increasing its density.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** gamma rays X rays ultraviolet visible light infrared radio. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key